

Step	Step Description	Agents Involved	Tool Calls and Parameters	Reason
1	Decompose query into steps	planning_agent	N/A	planning_agent chosen for its role in breaking down geospatial queries. Steps are: (1) Find Exeter area, (2) Search for houses in Exeter, (3) Apply conditions for hard to heat. This is based on the system prompt for planning_agent.
2	Find the area polygon for Exeter	named_area	call_os_ngd (polygon_or_point=True, filename='exeter_area_polygon')	named_area agent is used because the prompt specifies it provides polygons for named areas. No filters applied as per agent's system prompt.
3	Filter to the specific Exeter polygon	data_analysis_agent	generate_metadata_for_artifacts(['exeter_area_polygon']); code_executor (filter using name1_text, description, descriptiongroup for 'Exeter')	data_analysis_agent is used to filter the artifact to only polygons relevant to Exeter. Columns chosen based on metadata and agent prompt. Multiple columns used to increase search quality as per coding agent template.
4	Search for houses in the Exeter area	buildings	call_os_ngd (bbox='filtered_exeter_area_polygon', filename='all_buildings_in_filtered_exeter_area')	buildings agent is used as per its prompt for searching buildings within a bbox. The bbox is set to the filtered Exeter polygon artifact.
5	Filter for houses (residential, single address)	data_analysis_agent	generate_metadata_for_artifacts(['buildings_all_buildings_in_filtered_exeter_area']); code_executor (filter where buildinguse_addresscount_residential > 0 and buildinguse_addresscount_total = 1)	data_analysis_agent is used to filter for houses as defined in the buildings agent's special scenarios. Columns chosen based on metadata and system prompt.
6	Identify houses likely hard to heat	buildings, data_analysis_agent	send_message to data_analysis_agent (criteria: age pre-1999 or poor construction material); generate_metadata_for_artifacts(['houses_in_filtered_exeter_area']); code_executor (filter for buildingage_period in pre-1999 or constructionmaterial in poor types)	buildings agent delegates to data_analysis_agent for attribute-based filtering. Criteria and columns are based on the prompt and metadata. Multiple attempts to execute code due to variable naming errors, finally succeeded by passing the correct data structure.
7	Plot distribution of hard to heat houses	plotting_agent	generate_metadata_for_artifacts(['hard_to_heat_houses_in_exeter_area']); code_executor (plot folium map, convert datetime columns to string for serialization)	plotting_agent is used as per its template for geospatial visualization. CRS conversion and serialization fixes are applied based on folium/map requirements and error handling.
8	Summarize and communicate results	host_agent	N/A	host_agent summarizes the results for the user, referencing the criteria and the number of houses identified and mapped.

Overall Summary: - Databases used: Named area (for Exeter polygon), Buildings (for all buildings in Exeter), and subsequent filtered artifacts. - Filters/code: Filtering for Exeter used multiple columns (name, description). Houses were defined as buildings with residential address count > 0 and total address count = 1. Hard to heat houses were identified by age (pre-1999) or construction material (timber, non-standard, mixed, unknown, mobile home), as per metadata and system prompts. Folium was used for mapping, with serialization fixes for datetime columns. - Final outcome: 35,348 houses in Exeter were identified as likely hard to heat, and their distribution was visualized on an interactive map. The process involved careful stepwise filtering and attribute-based analysis, with all decisions grounded in agent prompts, metadata, and system policies.